Experiment No: 3

OBJECTIVES-

1. To understand the use of COMMIT, ROLLBACK, and SAVEPOINT commands.
2. To use ORDER BY and GROUP BY clauses.
3. To understand DCL commands (GRANT, REVOKE).

4.To implement JOIN operations in SQL.

Tables Used-

- TEACHER
- SALES
- DEPARTMENTS
- EMPLOYEES

DDL QUESTIONS (DATA DEFINITION LANGUAGE)

1. Create TEACHER Table

```
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary NUMBER
);
```

2. Create SALES Table

```
CREATE TABLE SALES (
  Sales_No INT PRIMARY KEY,
  Salesname VARCHAR(50),
  Branch VARCHAR(50),
  Salesamount NUMBER,
  DOB DATE
);
```

3. Create DEPARTMENTS Table

```
CREATE TABLE DEPARTMENTS (
   dept_no INT PRIMARY KEY,
   dept_name VARCHAR(50),
   dept_location VARCHAR(50)
);
```

4. Create EMPLOYEES Table

```
CREATE TABLE EMPLOYEES (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   emp_salary NUMBER,
   dept_no INT
);
```

DML QUESTIONS (DATA MANIPULATION LANGUAGE) FROM TEACHER TABLE

1. Insert Five Records

```
INSERT INTO TEACHER VALUES ('Amit Sharma',101,'2018-06-12','Mathematics','Delhi',40000);
INSERT INTO TEACHER VALUES ('Neha Verma',102,'2019-07-10','Commerce','Noida',38000);
INSERT INTO TEACHER VALUES ('Rohit Jain',101,'2020-01-15','Mathematics','Delhi',42000);
INSERT INTO TEACHER VALUES ('Pooja Mehta',103,'2017-03-18','Science','Gurgaon',45000);
INSERT INTO TEACHER VALUES ('Karan Singh',102,'2021-09-05','Commerce','Noida',36000);
```

2. Increment 25% Salary for Mathematics Department

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

3. Increment 15% Salary for Commerce Department

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

FROM SALES TABLE

1. Insert Five Records

```
INSERT INTO SALES VALUES (1,'Ravi','Delhi',50000,'1998-12-21');
INSERT INTO SALES VALUES (2,'Amit','Mumbai',42000,'1997-06-14');
INSERT INTO SALES VALUES (3,'Neha','Delhi',38000,'1999-12-05');
INSERT INTO SALES VALUES (4,'Karan','Chennai',46000,'1996-03-18');
INSERT INTO SALES VALUES (5,'Simran','Mumbai',52000,'1998-12-11');
```

2. Total Sales Amount in Each Branch

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

3. Average Sales Amount in Each Branch

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

4. Salesmen Born in December (Formatted Date)

```
SELECT Salesname,
    TO_CHAR(DOB, 'DD-Mon-YY') AS DOB
FROM SALES
WHERE TO_CHAR(DOB, 'MM') = '12';
```

5. Salesmen Ordered Alphabetically by Month

```
SELECT Salesname, DOB
FROM SALES
ORDER BY TO_CHAR(DOB, 'MM');
```

JOIN OPERATIONS

Insert Records into DEPARTMENTS

```
INSERT INTO DEPARTMENTS VALUES (10,'HR','Delhi');
INSERT INTO DEPARTMENTS VALUES (20,'Finance','Mumbai');
```

```
INSERT INTO DEPARTMENTS VALUES (30,'IT','Bangalore');
```

Insert Records into EMPLOYEES

```
INSERT INTO EMPLOYEES VALUES (1,'Rahul',50000,10);
INSERT INTO EMPLOYEES VALUES (2,'Neha',60000,20);
INSERT INTO EMPLOYEES VALUES (3,'Amit',45000,30);
INSERT INTO EMPLOYEES VALUES (4,'Pooja',55000,20);
INSERT INTO EMPLOYEES VALUES (5,'Rohit',40000,10);
```

1. Employee & Department Details (INNER JOIN)

```
SELECT emp_id, emp_name, emp_salary, dept_name, dept_location
FROM EMPLOYEES
INNER JOIN DEPARTMENTS
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;
```

2. LEFT OUTER JOIN

```
SELECT emp_name, dept_name
FROM EMPLOYEES
LEFT JOIN DEPARTMENTS
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;
```

3. RIGHT OUTER JOIN

```
SELECT emp_name, dept_name
FROM EMPLOYEES
RIGHT JOIN DEPARTMENTS
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;
```

4. Employees Earning More Than Average Salary

```
SELECT emp_name, emp_salary
FROM EMPLOYEES
WHERE emp_salary >
    (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

TCL QUESTIONS (TRANSACTION CONTROL LANGUAGE)

1. ROLLBACK on TEACHER Table

```
ROLLBACK;
```

➡️ Cancels all uncommitted changes.

2. COMMIT on TEACHER Table

COMMIT;

➡️ Saves all changes permanently.

3. SAVEPOINT Implementation

SAVEPOINT S1;

UPDATE TEACHER

SET Salary = Salary + 2000

WHERE DeptName = 'Science';

ROLLBACK TO S1;

DCL QUESTIONS (DATA CONTROL LANGUAGE)

1. Grant All Privileges on EMPLOYEES to DEPARTMENTS

GRANT ALL ON EMPLOYEES TO DEPARTMENTS;

2. Grant Selected Privileges

GRANT SELECT, INSERT ON EMPLOYEES TO DEPARTMENTS;

3. Revoke All Privileges

REVOKE ALL ON EMPLOYEES FROM DEPARTMENTS;

4. Revoke Selected Privileges

REVOKE INSERT ON EMPLOYEES FROM DEPARTMENTS;

Conclusion-

Thus, SQL commands for transaction control (COMMIT, ROLLBACK, SAVEPOINT), grouping and sorting, JOIN operations, and privilege management (GRANT, REVOKE) were successfully implemented and executed as per the given lab request.